

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

II Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2023

ME1003-1 – ELEMENTS OF MECHANICAL ENGINEERING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) A 4-cylinder 4-stroke engine running at 1000 rpm develops an indicated power of 15kW. The mean effective pressure is $5 \times 10^5 \text{ N/m}^2$. Find the diameter of the cylinder and the stroke of the piston when the ratio of diameter to stroke is 0.8.	5	L*3	3	2
	b) Explain the vapour absorption refrigeration system with a neat sketch.	5	L2	3	1
2.	a) Explain the working of four stroke C.I. engine with a neat sketch	6	L2	3	1
	b) Define the following: (i) Coefficient of Performance (ii) Tons of refrigeration.	4	L2	3	1
Unit – II					
3.	a) In a crossed belt drive the difference in tensions between the tight and slack sides of the belt is 1000 N. Find the tension on the slack and tight sides. If the angle of contact is 160° and its coefficient of friction is 0.3.	4	L3	4	2
	b) Explain different types of Gears with neat sketches.	6	L2	4	1
4.	a) Two spur gears A and B connect two parallel shafts that are 500 mm apart. Gear A runs at 400 rpm and gear B at 200 rpm. If the circular pitch is given to be 30 mm calculate the number of teeth on gears A and B.	5	L3	4	2
	b) Explain different types of belt drives with neat sketches. Also mention different materials used for making belts.	5	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
